

Kiyohiro (George) Nakayama
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OBJECTIVE

I am a second-year PhD student in Computer Science at Stanford University, advised by Professors Gordon Wetzstein and Leonidas Guibas. My research focuses on generative models for images, videos, and 3D content, with an emphasis on developing representations that enable efficient and scalable generation.

EDUCATION

Stanford University	Stanford, California, USA
<i>Ph.D Computer Science</i>	2025 -
Stanford University	Stanford, California, USA
<i>M.S. Computer Science</i>	2024 - 2025
Stanford University	Stanford, California, USA
<i>B.A. Mathematics</i>	2019 - 2024

GPA: 4.04/4.00

PUBLICATIONS

Level-of-Token Diffusion

Kiyohiro Nakayama, Brian Chao, Jan Ackermann, Hansheng Chen, Federico Tombari, Leonidas Guibas, Lior Yariv, Gordon Wetzstein
(In submission to ICLR 2027)

Garment Particles: A 2D–3D Symmetric Garment Representation for Generation and Editing

Kiyohiro Nakayama, I-Chao Shen, Ruofan Liu, Yiming Wang, Gordon Wetzstein, Takeo Igarashi
ACM **SIGGRAPH** Conference Papers, 2026

MeshFlow: Mesh Generation with Equivariant Flow Matching

Qi Sun, **Kiyohiro Nakayama**, Jing Nathan Yan, Qixing Huang, Alexander Rush, Leonidas Guibas, Gordon Wetzstein, Jing Liao, Guandao Yang
ACM **SIGGRAPH** Conference Papers, 2026

Image2Garment: Simulation-ready Garment Generation from a Single Image

Selim Emir Can*, Jan Ackermann*, **Kiyohiro Nakayama**, Ruofan Liu, Tong Wu, Yang Zheng, Hugo Bertiche, Menglei Chai, Thabo Beeler, Gordon Wetzstein
(In submission to 3DV 2027)

RBench-V: A Primary Assessment for Visual Reasoning Models with Multimodal Outputs

Meng-Hao Guo, Xuanyu Chu, Qianrui Yang, Zhe-Han Mo, Yiqing Shen, Pei-lin Li, Xinjie Lin, Jinnian Zhang, Xin-Sheng Chen, Yi Zhang, **Kiyohiro Nakayama**, Zhengyang Geng, Houwen Peng, Han Hu, Shi-min Hu
Advances in Neural Information Processing Systems (**NeurIPS**), Datasets and Benchmarks Track, 2025

GenAnalysis: Joint Shape Analysis by Learning Man-Made Shape Generators with Deformation Regularizations

Yuezhi Yang, Haitao Yang, **Kiyohiro Nakayama**, Xiangru Huang, Leonidas Guibas, Qixing Huang
ACM Transactions on Graphics (**SIGGRAPH**), 2025

RBench: Graduate-level Multi-disciplinary Benchmarks for LLM & MLLM Complex Reasoning Evaluation

Meng-Hao Guo, Jiajun Xu, Yi Zhang, Jiayi Song, Haoyang Peng, Yi-Xuan Deng, Xinzhi Dong, **Kiyohiro Nakayama**, Zhengyang Geng, Chen Wang, Bolin Ni, Guo-Wei Yang, Yongming Rao, Houwen Peng, Han Hu, Gordon Wetzstein, Shi-Min Hu

International Conference on Machine Learning (**ICML**), 2025

AIpparel: A Multimodal Foundation Model for Digital Garments

Kiyohiro Nakayama*, Jan Ackermann*, Timur Levent Kesdogan*, Yang Zheng, Maria Korosteleva, Olga Sorkine-Hornung, Leonidas J. Guibas, Guandao Yang, Gordon Wetzstein
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025

Towards Vision-Language-Garment Models for Web Knowledge Garment Understanding and Generation

Jan Ackermann, **Kiyohiro Nakayama**, Guandao Yang, Tong Wu, Gordon Wetzstein
MMFM Workshop at **CVPR**, 2025

ProvNeRF: Modeling per Point Provenance in NeRFs as a Stochastic Process
Kiyohiro Nakayama, Mikaela Angelina Uy, Yang You, Ke Li, Leonidas Guibas
Advances on Neural Information Processing Systems (**NeurIPS**), 2024

Semantic-Aware Transformation-Invariant RoI Align
Guo-Ye Yang, **Kiyohiro Nakayama**, Zi-Kai Xiao, Tai-Jiang Mu, Sharon Xiaolei Huang, Shi-Min Hu
AAAI Conference on Artificial Intelligence, 2024

NeRF Revisited: Fixing Quadrature Instability in Volume Rendering
Mikaela Angelina Uy, **Kiyohiro Nakayama**, Guandao Yang, Rahul Krishna Thomas, Leonidas Guibas, Ke Li
Advances on Neural Information Processing Systems (**NeurIPS**), 2023

DiffFacto: Controllable Part-Based 3D Point Cloud Generation with Cross Diffusion
Kiyohiro Nakayama, Mikaela Angelina Uy, Jiahui Huang, Shi-Min Hu, Ke Li, Leonidas Guibas
International Conference of Computer Vision (**ICCV**), 2023

EXPERIENCE

NVIDIA

Research Internship

California, USA
June 2026 – Present

- RL-based Polygon Packing
- Advisor: Nick Sharp, Mikaela Uy, and Donglai Xiang

University of Tokyo

Research Internship

Tokyo, Japan
July 2025 – September 2025

- Garment Representation for intuitive editing and design driven by a generative prior
- Advisor: Professor Takeo Igarashi and Ichao Shen

Tsinghua University

Research Internship

Beijing, China
January, 2022 – August, 2022

- 2D Image Segmentation with an attention-based, aspect-ratio aware feature extraction method.
- Advisor: Professor Shi-Min Hu.

INVITED TALKS

IGARASHI Laboratory

Host: Professor Takeo Igarashi

March 2024

Graphics and Geometric Computing Group at Tsinghua University

Host: Professor Shi-min Hu

December 2023

Stanford G-Cafe

April, 2023

References available upon request.